

edx Phot1x report template (2016/11)

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the date of receipt and acceptance should be inserted later

1 Introduction

The objective is to design an unbalanced MZI, extract waveguide group index through Free Spectral Range (FSR), which is the wavelength (or frequency) spacing between adjacent peaks in the MZI transmission spectrum. And to simulate different variations of unbalanced MZI, varying the difference between the path lengths (delta L), observing values to compare with real results. The device will be fabricated on 220nm SOI, TE polarized, using a 500nm strip waveguide.

2 Theory

2.1 Waveguide dispersion

Light propagation in a single-mode waveguide is characterized by the effective index $n_{\text{eff}}(\lambda)$. The group index, which determines the delay of optical signals and the interference fringe spacing, is

$$n_g(\lambda) = n_{\text{eff}}(\lambda) - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (1)$$

The effective index will be obtained from eigenmode simulations and fitted with a low-order polynomial to enable analytic evaluation of n_g .

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2.2 Mach–Zehnder interferometer

An unbalanced Mach–Zehnder interferometer (MZI) with path length difference $\Delta L = L_2 - L_1$ introduces a wavelength-dependent phase difference

$$\Delta\phi(\lambda) = \frac{2\pi n_g(\lambda) \Delta L}{\lambda}. \quad (2)$$

Assuming ideal 50:50 splitting and negligible loss, the normalized transmission is

$$T(\lambda) = \frac{1}{2} [1 + \cos(\Delta\phi(\lambda))]. \quad (3)$$

2.3 Free spectral range

Constructive interference occurs when $\Delta\phi = 2\pi m$. The wavelength spacing between adjacent maxima, the free spectral range (FSR), is

$$\text{FSR} \approx \frac{\lambda^2}{n_g \Delta L}. \quad (4)$$

2.4 Extraction of group index

From measured transmission spectra, the group index can be obtained directly as

$$n_g(\lambda) = \frac{\lambda^2}{\text{FSR} \cdot \Delta L}. \quad (5)$$

This equation forms the basis of the experiment: by fabricating MZIs with known ΔL and measuring the fringe spacing, the waveguide group index can be extracted and compared with simulated values.

3 Modelling and Simulation

3.1 Compact waveguide model

The effective index is represented by a polynomial around $\lambda_0 = 1550$ nm,

$$n_{\text{eff}}(\lambda) = a_0 + a_1(\lambda - \lambda_0) + a_2(\lambda - \lambda_0)^2, \quad (6)$$

with group index

$$n_g(\lambda) = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}. \quad (7)$$

Coefficients a_i will be obtained from eigenmode simulation of the 500 x 220 nm strip waveguide (TE).

3.2 Simulation results for waveguide

Waveguide simulations with Lumerical MODE FDE. Waveguide is 500nm in width, and 220nm in height. The effective index of the waveguide was simulated to be 2.46 at 1550nm wavelength.

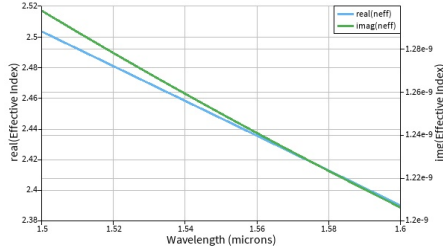


Fig. 1 Effective index

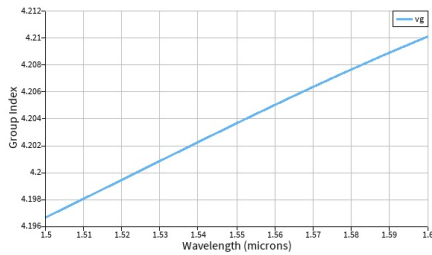


Fig. 2 Group index

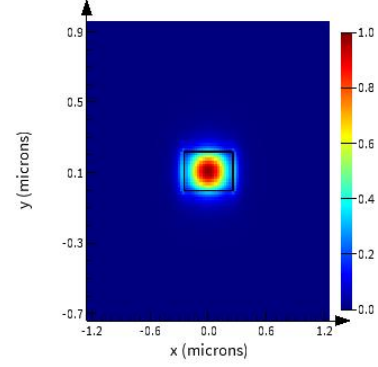


Fig. 3 Electric field intensity in Si waveguide

3.3 Interferometer model

The MZI transmission is

$$T(\lambda) = \frac{1}{2} \left[1 + \cos \left(\frac{2\pi n_g(\lambda) \Delta L}{\lambda} \right) \right]. \quad (8)$$

The expected free spectral range is

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L}. \quad (9)$$

3.4 Design sweep

Using the FSR equation in the previous section, and the group index of 4.19 (from lumerical MODE), these are the estimates for FSR. The Delta L values are derived from the GDS file.

ΔL (μm)	Expected FSR (nm)
19.0980	30.02nm
59.0780	9.71nm
99.8380	5.74 nm

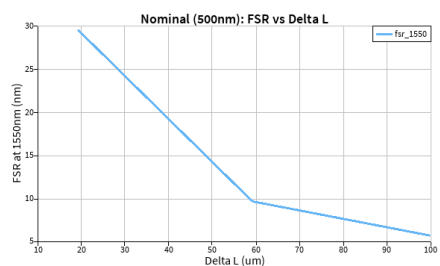
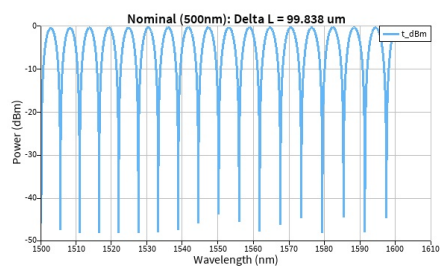
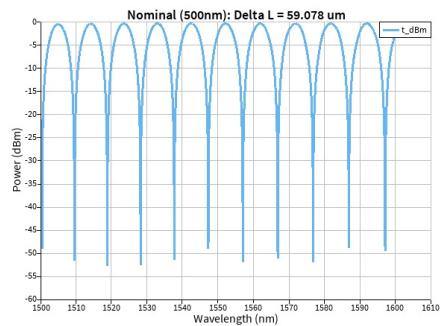
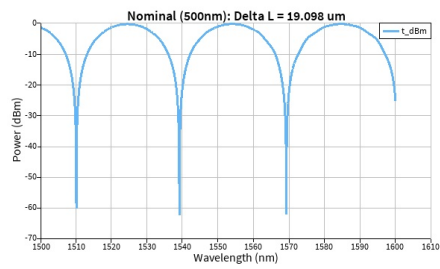
3.5 LUMERICAL INTERCONNECT simulation of unbalanced MZIs

The following are results from simulating the unbalanced MZI in interconnect. The waveguides and y branches in this simulation are from the EBeam design kit (ebeam_{wgintegral}1550).

Delta L: 19.098 μm , Max T: 0.932243, FSR at 1550nm: 29.5927 nm, ER: 62.1832 dB, ng: 4.251

Delta L: 59.078 μm , Max T: 0.932304, FSR at 1550nm: 9.67514 nm, ER: 52.3239 dB, ng: 4.2032

Delta L: 99.838 μm , Max T: 0.928265, FSR at 1550nm: 5.70449 nm, ER: 47.9462 dB, ng: 4.21843



3.6 Manufacturing variability, and corner study

Waveguide width ± 10 nm corner study in Lumerical Interconnect:

Corner: Width+10nm (510nm)

Delta L: 19.098 um, Max T: 0.930492, FSR at 1550nm: 29.8229 nm, ER: 60.5072 dB, ng: 4.21819

Delta L: 59.078 um, Max T: 0.930915, FSR at 1550nm: 9.64367 nm, ER: 52.9045 dB, ng: 4.21692

Delta L: 99.838 um, Max T: 0.929259, FSR at 1550nm: 5.76592 nm, ER: 48.0119 dB, ng: 4.17348

Corner: Width-10nm (490nm)

Delta L: 19.098 um, Max T: 0.932369, FSR at 1550nm: 29.1873 nm, ER: 61.6101 dB, ng: 4.31004

Delta L: 59.078 um, Max T: 0.931905, FSR at 1550nm: 9.54097 nm, ER: 52.4344 dB, ng: 4.26231

Delta L: 99.838 um, Max T: 0.929413, FSR at 1550nm: 5.70446 nm, ER: 48.0872 dB, ng: 4.21845

GROUP INDEX SUMMARY

Minimum ng: 4.17348 (Width+10nm (510nm) DL=99.838um)

Maximum ng: 4.31004 (Width-10nm (490nm) DL=19.098um)

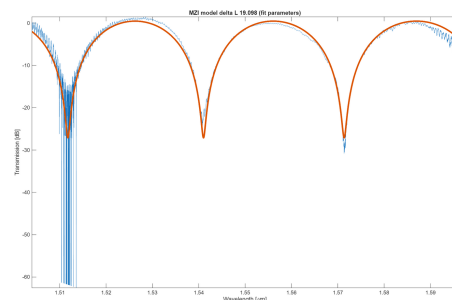
ng range: 0.136559

3.7 Measurement data, and verifying simulation results

The FittingMZI.mlx script seemed to work best for fitting the raw MZI data, so that was used as a base to then derive the FSR and ER.

For the MZI with delta L 19.098um:

FSR at 1550nm: 30.1288 nm. Extinction Ratio: 21.57 dB. ER from spectrum (max-min): 27.58 dB. ng at 1550nm: 4.17536. neff at 1550nm: 2.45109. Propagation loss (alpha): 0.0088 micron⁻¹ (760.95 dB/cm).dB/cm).



The FSR derived from the model can be seen in Fig. 5 below. Note that the simulations in Interconnect predicted an FSR value of 29.5927nm at the peak closest to 1550nm. The same FSR derived from the model of the data is 29.9010nm, this is about a 1% difference.

Model fit for delta L 59.078. FSR at 1550nm: 9.7100 nm. Extinction Ratio: 18.14 dB. ER from spectrum (max-min): 24.13 dB. ng at 1550nm: 4.18813. neff at 1550nm: 2.44914. Propagation loss (alpha): 0.0042 micron⁻¹ (366.04 dB/cm).

Model fit for delta L 99.838um. FSR at 1550nm: 5.7493 nm. Extinction Ratio: 35.35 dB. ER from spectrum (max-min): 41.37 dB. ng at 1550nm: 4.18552. neff at 1550nm: 2.45054. Propagation loss (alpha): 0.0003 micron⁻¹ (29.74 dB/cm).

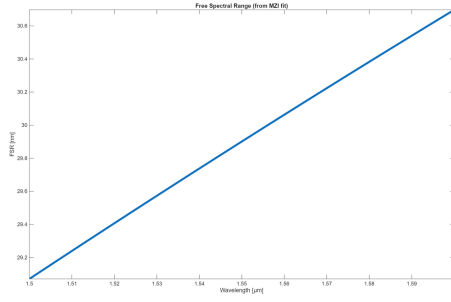


Fig. 4 FSR of MZI with Delta L 19.098 μm , derived from model based on raw data

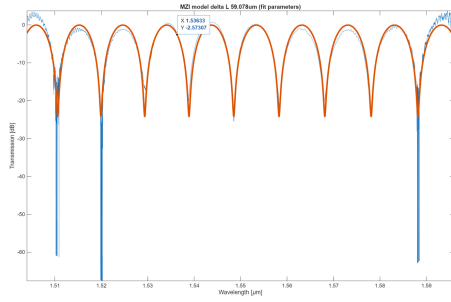


Fig. 5 This is a caption

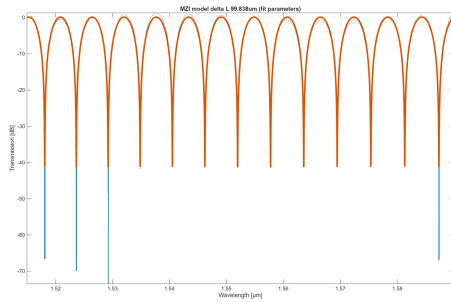


Fig. 6 This is a caption

4 Conclusion

The theory, the simulation results, and the derived values from the raw data largely matched.

Comparing the raw data taken from the chip in Fig. 8 below to the lumerical simulation in Fig. 7 below, the peak frequency looks about the same. There are differences in scaling of power due to the effects from the grating coupler being not fully captured by the simulation.

Various improvements can be made:

- Have waveguide height variations in the corner analysis.
- Account for waveguide bends in the simulation, currently all waveguides are treated as straight in the simulation.
- Use high fidelity FDTD 3d simulations on individual components to get better S Parameters for Interconnect.
- Account for temperature.
- Make use of the calibration structure to calibrate other results.
- Data cleaning, and removing noise would make fitting the polynomial to the data much easier.

I/We acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Enxiao Luan performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

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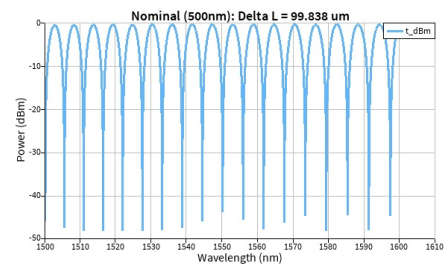


Fig. 7 Interconnect simulation for MZI delta L 99.838um

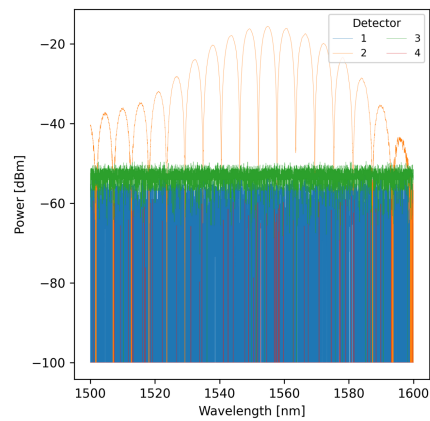


Fig. 8 Data for MZI delta L 99.838um